

Development Activity 3 — Parallel Module Development and Integration

Activity Objective

In the previous activities, you developed the frontend prototype of your semester project and introduced different user roles and role-based interfaces.

In this activity, you will extend the same project by developing **two separate functional modules simultaneously**. Each group member will be responsible for one module. After completing their individual tasks, both members will integrate their modules into the existing project.

Important: Continue working on your existing semester project. Do not create a separate practice application. This activity can be completed locally using HTML, CSS, and JavaScript, without an internet connection or backend server.

Part 1 — Select Two Complementary Modules

As a group, select two modules that are relevant to your project. The modules should perform different but related functions.

Examples:

Semester Project	Module 1	Module 2
Provincial Complaint Management System	Submit Complaint	View and Track Complaints
Alumni Collaboration System	Create Alumni Profile	Browse Alumni Directory
Student Community Management System	Create Community Post	View and Search Posts
Blood Donor Management System	Register Donor	Search Available Donors
Internship Management System	Post Internship	Browse Internship Opportunities
Food Donation Management System	Register Donation	View Available Donations
Scholarship Assistance System	Submit Scholarship Application	View Application Status
Lost & Found System	Report Lost Item	Browse Reported Items

You may choose different modules according to your project requirements.

Project Name: _____

Module 1: _____

Module 2: _____

Define the responsibilities

Briefly describe what each module will do.

Module 1:

Module 2:

Part 2 — Member 1: Develop Module 1

Responsible Group Member: _____

Develop the first module independently using HTML, CSS, and JavaScript.

Your module must include the following:

1. **User Interface:** Create a clean interface appropriate for your selected module.
2. **Input or Action:** Allow the user to perform the module's primary task, such as submitting a complaint, creating a profile, or posting an internship.
3. **Input Validation:** Check that required information is provided before accepting the submission.
4. **JavaScript Functionality:** Make the primary action functional rather than leaving it as a decorative button.
5. **Feedback:** Display an appropriate success or error message after the action.
6. **Data Display:** Display the submitted item or updated information on the page. You may use JavaScript arrays and sample records; a database is not required for this activity.

Example

For a Provincial Complaint Management System, Module 1 could allow a citizen to:

- Enter a complaint title.
- Select a complaint category.
- Provide a description.
- Submit the complaint.
- See a confirmation message and the newly added complaint.

Your module's primary function:

Input validation rule:

JavaScript interaction implemented:

Part 3 — Member 2: Develop Module 2

Responsible Group Member: _____

Develop the second module independently using HTML, CSS, and JavaScript.

Your module must include the following:

1. **User Interface:** Create a clean interface appropriate for your selected module.
2. **Primary Functionality:** Implement the module's main task.
3. **Input Validation:** If the module accepts input, validate the required information.
4. **JavaScript Functionality:** Implement at least one meaningful interaction.
5. **Feedback:** Provide suitable feedback when the user completes an action or provides invalid input.
6. **Data Display:** Display relevant information using sample records or locally maintained JavaScript data.

Example

For the same Provincial Complaint Management System, Module 2 could allow an officer or citizen to:

- View a list of sample complaints.
- Search for a complaint by its reference number.
- Filter complaints by status.
- View complaint details.

Your module's primary function:

Input validation or search rule:

JavaScript interaction implemented:

Part 4 — Integrate Both Modules

After completing the individual modules, work together to incorporate them into your existing semester project.

Both members are responsible for ensuring that the final prototype works as one application.

Complete the following tasks:

- Place both modules within the same project folder.
- Link both modules to the existing project dashboard or navigation menu.
- Ensure that users can move between the main interface and both modules.
- Use consistent headings, navigation, buttons, fonts, and overall styling.
- Check that file names and local links are correct.
- Remove broken links and buttons that do not perform their intended actions.

Integration Checklist

- Module 1 opens and works correctly.
- Module 2 opens and works correctly.
- Both modules are accessible through the project interface.
- Navigation between pages works locally.
- Both modules follow a consistent design.
- Existing functionality from the previous activities still works.

Note: A backend or permanent database connection is not required. Sample data and JavaScript arrays are sufficient. Data does not have to persist after refreshing the page.

Part 5 — Test Both Modules

Each group member must test their own module, and both members must test the integrated project.

For **each module**, perform these checks:

Test Case	Expected Result	Actual Result
Submit or perform a valid action	The action succeeds	_____
Leave a required field empty	The system prevents submission or displays an error	_____
Perform a second action, such as searching or filtering	The interface updates appropriately	_____

Then test the combined project.

Module 1 issue discovered:

Module 2 issue discovered:

Correction made:

Part 6 — Security-Aware Development

Although this activity focuses on frontend functionality, consider how each module may be misused.

For each module, identify one possible problem and a suitable response.

Module	Possible Misuse or Problem	Proposed Response
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Module 1	_____	_____
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Module 2	_____	_____
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Examples include submitting incomplete information, attempting to access information that should remain private, or entering unexpected values.

At this stage, document the concern and implement suitable frontend validation where appropriate. Remember that frontend validation alone cannot enforce security in a real application; backend enforcement will be addressed as the semester progresses.

Submission Requirements

Submit the following information:

Project Name: _____

Group Members: _____

Module 1: _____

Module 2: _____

Module 1 functionality implemented:

Module 2 functionality implemented:

JavaScript interactions implemented:

1. _____
2. _____

One problem discovered during testing:

How it was corrected:

Deliverables

1. Updated project source code, including HTML, CSS, and JavaScript files.
2. Both functional modules integrated into the existing project.
3. Screenshots showing each module.
4. Completed testing and security-awareness notes.

Keep the project files together in one folder so that the integrated prototype can be opened and tested locally without an internet connection.